

CV Equations

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Legend

- State 1:
 - Three inlets/injectors: 2 smaller ones for air and 1 for hydrogen
 - Subscript “A” stands for air
 - Subscript “B” stands for hydrogen
 - ρ (kg/m^3) - density
 - u (m/s) – velocity
 - A_A (m^2) - area of the two air injectors
 - A_B (m^2) - area of the hydrogen injector
 - h (J/kg) – static enthalpy – it is a function of T
 - $\dot{m}_A$ (kg/s) - mass flow rate out of air injectors
 - $\dot{m}_B$ (kg/s) - mass flow rate out of hydrogen injectors
- State 2:
 - One outlet (the inlet of the preburner)
 - Subscript “i” stands for
 - ρ (kg/m^3) - density
 - u (m/s) – velocity
 - A_i (m^2) – area of preburner inlet
 - h (J/kg) – static enthalpy – it is a function of T
 - $\dot{m}_i$ (kg/s) - mass flow rate at preburner inlet

Main equations

- Mass Conservation

- $\dot{m}_A + \dot{m}_B = \dot{m}_i$
- $\dot{m}_A + \dot{m}_B - \dot{m}_i = 0$
- $\rho_A * u_A * A_A + \rho_B * u_B * A_B - \rho_i * u_i * A_i = 0$

- Momentum Conservation

- $\int_{CS} \rho * u * u * dA = \sum F$
- $\dot{m}_i * u_i - \sum_{Inlets} (\dot{m} * u) = \sum F$

- Energy Conservation

- $\int_{CS} \rho * \left(h + \frac{u^2}{2} \right) * u * dA = \dot{Q} - \dot{W}$
- Assumptions: Adiabatic ($\dot{Q} = 0$), no shaft work ($\dot{W} = 0$)
- $\int_{CS} \rho * \left(h + \frac{u^2}{2} \right) * u * dA = 0$
- $\dot{m}_i * \left(h_i + \frac{u_i^2}{2} \right) - \sum_{Inlets} (\dot{m} * \left(h + \frac{u^2}{2} \right)) = 0$

Momentum Conservation Equation

- $\dot{m}_i * u_i - \sum_{Inlets} (\dot{m} * u) = \sum F$
- $\dot{m}_i * u_i - \sum_{Inlets} (\dot{m} * u) = (P_A * A_A + P_B * A_B) - P_i * A_i$
- $\dot{m}_i * u_i - \sum_{Inlets} (\dot{m} * u) - (P_A * A_A + P_B * A_B) + P_i * A_i = 0$
- $\dot{m}_i * u_i - (\dot{m}_A * u_A + \dot{m}_B * u_B) - (P_A * A_A + P_B * A_B) + P_i * A_i = 0$

Energy Conservation Equation

- $\int_{CS} \rho * \left(h_{static} + \frac{u^2}{2} \right) * u * dA = 0$
- $\dot{m}_i * \left(h_i + \frac{u_i^2}{2} \right) - \sum_{Inlets} (\dot{m} * \left(h + \frac{u^2}{2} \right)) = 0$
- $\dot{m}_i * \left(h_i + \frac{u_i^2}{2} \right) - [(\dot{m}_A * \left(h_A + \frac{u_A^2}{2} \right)) + (\dot{m}_B * \left(h_B + \frac{u_B^2}{2} \right))] = 0$

Equation List

- $F_1 = \rho_A * u_A * A_A + \rho_B * u_B * A_B - \rho_i * u_i * A_i = 0$
- $F_2 = \dot{m}_i * u_i - (\dot{m}_A * u_A + \dot{m}_B * u_B) - (P_A * A_A + P_B * A_B) + P_i * A_i = 0$
- $F_3 = \dot{m}_i * (h_i + \frac{u_i^2}{2}) - [(\dot{m}_A * (h_A + \frac{u_A^2}{2})) + (\dot{m}_B * (h_B + \frac{u_B^2}{2}))] = 0$

Simplifying: From three equations three unknowns to two equations two unknowns.

Part 1:

- $F_1 = \rho_A * u_A * A_A + \rho_B * u_B * A_B = \rho_i * u_i * A_i$
- $F_1 = \dot{m}_A + \dot{m}_B = \dot{m}_i$
- $\frac{F_2}{F_1} = \frac{\dot{m}_i * u_i - (\dot{m}_A * u_A + \dot{m}_B * u_B) - (P_A * A_A + P_B * A_B) + P_i * A_i}{\dot{m}_i} = \frac{0}{\dot{m}_A + \dot{m}_B}$
- $\frac{F_2}{F_1} = u_i - \frac{\dot{m}_A}{\dot{m}_i} * u_A - \frac{\dot{m}_B}{\dot{m}_i} * u_B - \frac{P_A * A_A}{\dot{m}_i} - \frac{P_B * A_B}{\dot{m}_i} + \frac{P_i * A_i}{\dot{m}_i} = 0$
- $P_A = \rho_A * R * T_A, P_B = \rho_B * R * T_B, P_i = \rho_i * R * T_i$
- $\frac{F_2}{F_1} = u_i - \frac{\dot{m}_A}{\dot{m}_i} * u_A - \frac{\dot{m}_B}{\dot{m}_i} * u_B - \frac{\rho_A * R * T_A * A_A}{\dot{m}_i} - \frac{\rho_B * R * T_B * A_B}{\dot{m}_i} + \frac{\rho_i * R * T_i * A_i}{\dot{m}_i} = 0$
- $\rho_A = \frac{\dot{m}_A}{u_A * A_A}, \rho_B = \frac{\dot{m}_B}{u_B * A_B}, \rho_i = \frac{\dot{m}_i}{u_i * A_i}$
- $\frac{F_2}{F_1} = u_i - \frac{\dot{m}_A}{\dot{m}_i} * u_A - \frac{\dot{m}_B}{\dot{m}_i} * u_B - \frac{\frac{\dot{m}_A}{u_A * A_A} * R * T_A * A_A}{\dot{m}_i} - \frac{\frac{\dot{m}_B}{u_B * A_B} * R * T_B * A_B}{\dot{m}_i} + \frac{\frac{\dot{m}_i}{u_i * A_i} * R * T_i * A_i}{\dot{m}_i} = 0$
- $\frac{F_2}{F_1} = u_i - \frac{\dot{m}_A}{\dot{m}_i} * u_A - \frac{\dot{m}_B}{\dot{m}_i} * u_B - \frac{\dot{m}_A * R * T_A}{\dot{m}_i * u_A} - \frac{\dot{m}_B * R * T_B}{\dot{m}_i * u_B} + \frac{R * T_i}{u_i} = 0$
- $E_1 = u_i - \frac{\dot{m}_A}{\dot{m}_i} * u_A - \frac{\dot{m}_B}{\dot{m}_i} * u_B - \frac{\dot{m}_A * R * T_A}{\dot{m}_i * u_A} - \frac{\dot{m}_B * R * T_B}{\dot{m}_i * u_B} + \frac{R * T_i}{u_i} = 0$

Simplifying: From three equations three unknowns to two equations two unknowns.

Part 2:

- $F_1 = \rho_A * u_A * A_A + \rho_B * u_B * A_B = \rho_i * u_i * A_i$

- $F_1 = \dot{m}_A + \dot{m}_B = \dot{m}_i$

- $\frac{F_3}{F_1} = \frac{\dot{m}_i * (h_i + \frac{u_i^2}{2}) - [(\dot{m}_A * (h_A + \frac{u_A^2}{2})) + (\dot{m}_B * (h_B + \frac{u_B^2}{2}))]}{\dot{m}_i} = \frac{0}{\dot{m}_A + \dot{m}_B}$

- $\frac{F_3}{F_1} = (h_i + \frac{u_i^2}{2}) - \frac{\dot{m}_A}{\dot{m}_i} (h_A + \frac{u_A^2}{2}) - \frac{\dot{m}_B}{\dot{m}_i} (h_B + \frac{u_B^2}{2}) = 0$

- $E_2 = (h_i + \frac{u_i^2}{2}) - \frac{\dot{m}_A}{\dot{m}_i} (h_A + \frac{u_A^2}{2}) - \frac{\dot{m}_B}{\dot{m}_i} (h_B + \frac{u_B^2}{2}) = 0$

Final Equations:

- $E_1 = u_i - \frac{\dot{m}_A}{\dot{m}_i} * u_A - \frac{\dot{m}_B}{\dot{m}_i} * u_B - \frac{\dot{m}_A * R * T_A}{\dot{m}_i * u_A} - \frac{\dot{m}_B * R * T_B}{\dot{m}_i * u_B} + \frac{R * T_i}{u_i} = 0$
- $E_2 = (h_i + \frac{u_i^2}{2}) - \frac{\dot{m}_A}{\dot{m}_i} (h_A + \frac{u_A^2}{2}) - \frac{\dot{m}_B}{\dot{m}_i} (h_B + \frac{u_B^2}{2}) = 0$

Entropy Equations

- $\dot{S}_{gen} = \dot{m}_i * s_i - \sum_{species} \dot{m}_{species} * s_{species}$
- $\dot{S}_{gen} = \dot{m}_i * s_i - (\dot{m}_A * s_A + \dot{m}_B * s_B)$
 - Will be getting entropy values from cantera based off of P,T,X values

Entropy Gen to Stag Pressure

- $\dot{S}_{gen} = c_p(T) * \ln \frac{T_{o2}}{T_{o1}} - R * \ln \frac{P_{o2}}{P_{o1}}$
 - Mixing CV is adiabatic so $T_{o2} = T_{o1}$
- $\dot{S}_{gen} = -R * \ln \frac{P_{o2}}{P_{o1}}$
- $e^{-\frac{\dot{S}_{gen}}{R}} = \frac{P_{o2}}{P_{o1}}$
- $P_{o2} = e^{-\frac{\dot{S}_{gen}}{R}} * P_{o1}$
- $P_{o1} = \frac{\sum_{species} \dot{m}_{species} * P_{o,species}}{\sum_{species} \dot{m}_{species}}$
- $P_{o1} = \frac{\dot{m}_A * P_{o,A} + \dot{m}_B * P_{o,B}}{\dot{m}_A + \dot{m}_B}$
- $P_{o2} = e^{-\frac{\dot{S}_{gen}}{R}} * \frac{\dot{m}_A * P_{o,A} + \dot{m}_B * P_{o,B}}{\dot{m}_A + \dot{m}_B}$

Justification For E3

- I am basically trying to create an equation in which I will be able to solve for static variables based off of a given/known stag pressure
- It is basically a balance between mass flow rate and stagnation values

Solving for static var after adjusting P_{o2}

- $\dot{m}_i = \rho_i * u_i * A_i$
- $P_i = \rho_i * R * T_i$
- $T_{oi} = T_i \left(1 + \frac{\gamma(T)-1}{2} * M_i^2 \right)$
- $T_i = \frac{T_{oi}}{\left(1 + \frac{\gamma(T)-1}{2} * M_i^2 \right)}$
- $P_{oi} = P_i \left(1 + \frac{\gamma(T)-1}{2} * M_i^2 \right)^{\frac{\gamma(T)}{\gamma(T)-1}}$
- $P_i = \frac{P_{oi}}{\left(1 + \frac{\gamma(T)-1}{2} * M_i^2 \right)^{\frac{\gamma(T)}{\gamma(T)-1}}}$

Solving for static var after adjusting P_{o2} cont.

- $\rho_i = \frac{P_i}{R * T_i}$
- $\rho_i = \frac{P_{oi}}{\left(1 + \frac{\gamma(T)-1}{2} * M_i^2\right)^{\frac{\gamma(T)}{\gamma(T)-1}}} / \frac{R * T_{oi}}{\left(1 + \frac{\gamma(T)-1}{2} * M_i^2\right)}$
- $\rho_i = \frac{P_{oi}}{R * T_{oi}} * \frac{\left(1 + \frac{\gamma(T)-1}{2} * M_i^2\right)^{\frac{\gamma(T)}{\gamma(T)-1}}}{\left(1 + \frac{\gamma(T)-1}{2} * M_i^2\right)^{\frac{\gamma(T)}{\gamma(T)-1}}}$
- $\rho_i = \frac{P_{oi}}{R * T_{oi}} * \left(1 + \frac{\gamma(T)-1}{2} * M_i^2\right) * \left(1 + \frac{\gamma(T)-1}{2} * M_i^2\right)^{-1 * \frac{\gamma(T)}{\gamma(T)-1}}$
- $\rho_i = \frac{P_{oi}}{R * T_{oi}} * \left(1 + \frac{\gamma(T)-1}{2} * \frac{u_i^2}{\sqrt{\gamma(T) * R * T_i}}\right)^{1 + (-1 * \frac{\gamma(T)}{\gamma(T)-1})}$
- $\rho_i = \frac{P_{oi}}{R * T_{oi}} * \left(1 + \frac{\gamma(T)-1}{2} * \frac{u_i^2}{\gamma(T) * R * T_i}\right)^{1 + (-1 * \frac{\gamma(T)}{\gamma(T)-1})}$

Solving for static var after adjusting P_{o2} cont.

- $\dot{m}_i = \frac{P_{oi}}{R * T_{oi}} * \left(1 + \frac{\gamma(T_i) - 1}{2} * \frac{u_i^2}{\gamma(T_i) * R * T_i} \right)^{1 + \left(-1 * \frac{\gamma(T_i)}{\gamma(T_i) - 1} \right)} * u_i * A_i$
- $E_3 = \frac{P_{oi}}{R * T_{oi}} * \left(1 + \frac{\gamma(T_i) - 1}{2} * \frac{u_i^2}{\gamma(T_i) * R * T_i} \right)^{1 + \left(-1 * \frac{\gamma(T_i)}{\gamma(T_i) - 1} \right)} * u_i * A_i - \dot{m}_i$

Equations I am using for solving the static var after adjusting P_{o2}

- $E_1 = u_i - \frac{\dot{m}_A}{\dot{m}_i} * u_A - \frac{\dot{m}_B}{\dot{m}_i} * u_B - \frac{\dot{m}_A * R * T_A}{\dot{m}_i * u_A} - \frac{\dot{m}_B * R * T_B}{\dot{m}_i * u_B} + \frac{R * T_i}{u_i} = 0$
- $E_3 = \frac{P_{oi}}{R * T_{oi}} * \left(1 + \frac{\gamma(T_i) - 1}{2} * \frac{u_i^2}{\gamma(T) * R * T_i} \right)^{1 + \left(-1 * \frac{\gamma(T_i)}{\gamma(T_i) - 1} \right)} * u_i * A_i - \dot{m}_i$
- 2eq, 2 unknowns (u_i, T_i)

Justification for E4

- E4 is basically the same as E2 but rewritten with stag temp

Total Energy Equation

- $T_i = \frac{T_{oi}}{\left(1 + \frac{\gamma(T_i)-1}{2} * M_i^2\right)}$
- $T_{oi} = T_i \left(1 + \frac{\gamma(T_i)-1}{2} * M_i^2\right)$
- $h_{oi}(T) = c_p(T) * T_{oi} = h(T_i) + \frac{u_i^2}{2}$ using nasa polynomials for h(T)
- $E_4 = h(T_i) + \frac{u_i^2}{2} - c_p(T_i) * T_{oi}$
 - T_{oi} is known
 - Just want to include it because mixing is adiabatic, so I believe it is like another way of constraining the actual value of total enthalpy